

# KIT SAFETY DATA SHEET



**Kit Product Name** SEQuoia Complete Stranded RNA Library Prep Kit

**Kit Catalog Number(s)** 17005710, 17005726

**Revision date** 19-Mar-2021

## Kit Contents

| Catalog Number(s)                         | Product Name                                    |
|-------------------------------------------|-------------------------------------------------|
| 12011293, 12011942                        | SEQuoia Complete Reagent F Amplification Buffer |
| 12011303, 12011945                        | SEQuoia Complete Reagent B End Repair Enzyme    |
| 12011302, 12011943                        | SEQuoia Complete Reagent A Fragmentation Buffer |
| 12011304, 12011946                        | SEQuoia Complete Reagent C Poly(A) Buffer       |
| 12011291, 12011944                        | SEQuoia Complete Reagent D Poly(A) Polymerase   |
| 12011292, 12011952                        | SEQuoia Complete Reagent E SEQzyme Mix          |
| 12011294, 12011941, 12012096,<br>12012097 | SEQuoia Complete Purification Beads             |



# SAFETY DATA SHEET

Revision date 23-Dec-2020

Revision Number 1

## 1. Identification

### Product identifier

**Product Name** SEQuoia Complete Reagent F Amplification Buffer

### Other means of identification

**Catalog Number(s)** 12011293, 12011942

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

**Restrictions on use** No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories (Canada) Ltd.  
1329 Meyerside Drive  
Mississauga, ON L5T 1C9  
Canada

#### **Technical Service**

1-800-361-1808  
support@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC Canada:1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

Specific target organ toxicity (single exposure)

Category 1

### Label elements

#### **Danger**

#### **Hazard statements**

Causes damage to organs



#### **Precautionary Statements - Prevention**

Do not breathe dust/fume/gas/mist/vapors/spray

Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product

**Precautionary Statements - Response**

IF exposed or concerned: Call a POISON CENTER or doctor

**Precautionary Statements - Disposal**

Dispose of contents/container to an approved waste disposal plant

**Other information**

Causes mild skin irritation Harmful to aquatic life with long lasting effects Harmful to aquatic life

**3. Composition/information on ingredients****Substance**

Not applicable.

**Mixture**

| Chemical name                             | CAS No  | Weight-% | Hazardous Material<br>Information Review Act<br>registry number<br>(HMIRA registry #) | Date HMIRA filed and<br>date exemption<br>granted (if applicable) |
|-------------------------------------------|---------|----------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| 1,2,3-Propanetriol                        | 56-81-5 | 10 - 30  | -                                                                                     |                                                                   |
| Methanaminium, N,N,N-trimethyl-, chloride | 75-57-0 | 1 - 5    | -                                                                                     |                                                                   |

**4. First-aid measures****Description of first aid measures**

|                       |                                                                                                                                                               |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b> | Show this safety data sheet to the doctor in attendance.                                                                                                      |
| <b>Inhalation</b>     | Remove to fresh air. IF exposed or concerned: Get medical advice/attention.                                                                                   |
| <b>Eye contact</b>    | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. If symptoms persist, call a physician.                               |
| <b>Skin contact</b>   | Wash off immediately with plenty of water for at least 15 minutes. If symptoms persist, call a physician.                                                     |
| <b>Ingestion</b>      | Do NOT induce vomiting. Clean mouth with water and drink afterwards plenty of water. Never give anything by mouth to an unconscious person. Call a physician. |

**Most important symptoms and effects, both acute and delayed**

|                 |                                                     |
|-----------------|-----------------------------------------------------|
| <b>Symptoms</b> | Prolonged contact may cause redness and irritation. |
|-----------------|-----------------------------------------------------|

**Indication of any immediate medical attention and special treatment needed**

|                           |                        |
|---------------------------|------------------------|
| <b>Note to physicians</b> | Treat symptomatically. |
|---------------------------|------------------------|

**5. Fire-fighting measures**

|                                     |                                                                                |
|-------------------------------------|--------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b> | Use extinguishing measures that are appropriate to local circumstances and the |
|-------------------------------------|--------------------------------------------------------------------------------|

surrounding environment.

**Unsuitable extinguishing media** No information available.

**Specific hazards arising from the chemical** None known.

**Explosion data**

**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** None.

**Special protective equipment for fire-fighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

## 6. Accidental release measures

### Personal precautions, protective equipment and emergency procedures

**Personal precautions** Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas.

**Other information** Refer to protective measures listed in Sections 7 and 8.

### Methods and material for containment and cleaning up

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labeled containers.

## 7. Handling and storage

### Precautions for safe handling

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

### Conditions for safe storage, including any incompatibilities

**Storage Conditions** Store locked up. Store according to product and label instructions.

## 8. Exposure controls/personal protection

### Control parameters

#### Exposure Limits

| Chemical name                 | Alberta                   | British Columbia                                      | Ontario | Quebec                    |
|-------------------------------|---------------------------|-------------------------------------------------------|---------|---------------------------|
| 1,2,3-Propanetriol<br>56-81-5 | TWA: 10 mg/m <sup>3</sup> | TWA: 10 mg/m <sup>3</sup><br>TWA: 3 mg/m <sup>3</sup> |         | TWA: 10 mg/m <sup>3</sup> |

### Appropriate engineering controls

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

### Individual protection measures, such as personal protective equipment

**Eye/face protection** Wear safety glasses with side shields (or goggles).

|                                |                                                                                                                                                                          |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hand protection                | Wear suitable gloves.                                                                                                                                                    |
| Skin and body protection       | Wear suitable protective clothing.                                                                                                                                       |
| Respiratory protection         | No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required. |
| General hygiene considerations | Handle in accordance with good industrial hygiene and safety practice.                                                                                                   |

## 9. Physical and chemical properties

### Information on basic physical and chemical properties

|                |                          |
|----------------|--------------------------|
| Physical state | Liquid                   |
| Appearance     | aqueous solution         |
| Color          | colorless                |
| Odor           | Odorless                 |
| Odor threshold | No information available |

| Property                               | Values            | Remarks • Method |
|----------------------------------------|-------------------|------------------|
| pH                                     | 8-9               |                  |
| Melting point / freezing point         | No data available | None known       |
| Boiling point / boiling range          | No data available | None known       |
| Flash point                            | No data available | None known       |
| Evaporation rate                       | No data available | None known       |
| Flammability (solid, gas)              | No data available | None known       |
| Flammability Limit in Air              |                   | None known       |
| Upper flammability or explosive limits | No data available |                  |
| Lower flammability or explosive limits | No data available |                  |
| Vapor pressure                         | No data available | None known       |
| Vapor density                          | No data available | None known       |
| Relative density                       | No data available | None known       |
| Water solubility                       | Miscible in water |                  |
| Solubility in other solvents           | No data available | None known       |
| Partition coefficient                  | No data available | None known       |
| Autoignition temperature               | No data available | None known       |
| Decomposition temperature              |                   | None known       |
| Kinematic viscosity                    | No data available | None known       |
| Dynamic viscosity                      | No data available | None known       |

### Other information

|                      |                 |
|----------------------|-----------------|
| Explosive properties | Not applicable. |
| Oxidizing properties | Not applicable. |
| Softening point      | Not applicable  |
| Molecular weight     | Not applicable  |
| VOC Content (%)      | Not applicable  |

## 10. Stability and reactivity

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Reactivity                         | No information available.                 |
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |

**Hazardous decomposition products** None known based on information supplied.

## 11. Toxicological information

### Information on likely routes of exposure

#### Product Information

|              |                                                                                                |
|--------------|------------------------------------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available.                              |
| Eye contact  | Specific test data for the substance or mixture is not available.                              |
| Skin contact | Specific test data for the substance or mixture is not available. Causes mild skin irritation. |
| Ingestion    | Specific test data for the substance or mixture is not available.                              |

### Symptoms related to the physical, chemical and toxicological characteristics

**Symptoms** Prolonged contact may cause redness and irritation.

### Acute toxicity

#### Numerical measures of toxicity

The following values are calculated based on chapter 3.1 of the GHS document .

#### Component Information

| Chemical name                                           | Oral LD50             | Dermal LD50          | Inhalation LC50                     |
|---------------------------------------------------------|-----------------------|----------------------|-------------------------------------|
| 1,2,3-Propanetriol<br>56-81-5                           | = 12600 mg/kg ( Rat ) | > 10 g/kg ( Rabbit ) | > 570 mg/m <sup>3</sup> ( Rat ) 1 h |
| Methanaminium,<br>N,N,N-trimethyl-, chloride<br>75-57-0 | = 50 mg/kg ( Rat )    | -                    | -                                   |

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                                          |                                                                                                                                                                                                                                                                                          |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin corrosion/irritation</b>         | Classification based on data available for ingredients. May cause skin irritation.                                                                                                                                                                                                       |
| <b>Serious eye damage/eye irritation</b> | Based on available data, the classification criteria are not met.                                                                                                                                                                                                                        |
| <b>Respiratory or skin sensitization</b> | Based on available data, the classification criteria are not met.                                                                                                                                                                                                                        |
| <b>Germ cell mutagenicity</b>            | Based on available data, the classification criteria are not met.                                                                                                                                                                                                                        |
| <b>Carcinogenicity</b>                   | Based on available data, the classification criteria are not met.                                                                                                                                                                                                                        |
| <b>Reproductive toxicity</b>             | Based on available data, the classification criteria are not met.                                                                                                                                                                                                                        |
| <b>STOT - single exposure</b>            | Based on the classification criteria of the Globally Harmonized System as adopted in the country or region with which this safety data sheet complies, this product has been determined to cause systemic target organ toxicity from acute exposure. (STOT SE). Causes damage to organs. |
| <b>STOT - repeated exposure</b>          | Based on available data, the classification criteria are not met.                                                                                                                                                                                                                        |
| <b>Target organ effects</b>              | Kidney, Respiratory system, Eyes, Skin.                                                                                                                                                                                                                                                  |
| <b>Aspiration hazard</b>                 | Based on available data, the classification criteria are not met.                                                                                                                                                                                                                        |

## 12. Ecological information

**Ecotoxicity** Harmful to aquatic life with long lasting effects.

| Chemical name                                           | Algae/aquatic plants | Fish                                                   | Toxicity to microorganisms | Crustacea                                   |
|---------------------------------------------------------|----------------------|--------------------------------------------------------|----------------------------|---------------------------------------------|
| 1,2,3-Propanetriol<br>56-81-5                           | -                    | LC50: 51 - 57mL/L (96h, <i>Oncorhynchus mykiss</i> )   | -                          | EC50: >500mg/L (24h, <i>Daphnia magna</i> ) |
| Methanaminium,<br>N,N,N-trimethyl-, chloride<br>75-57-0 | -                    | LC50: 431 - 495mg/L (96h, <i>Pimephales promelas</i> ) | -                          | -                                           |

**Persistence and degradability** No information available.

**Bioaccumulation** There is no data for this product.

#### Component Information

| Chemical name                 | Partition coefficient |
|-------------------------------|-----------------------|
| 1,2,3-Propanetriol<br>56-81-5 | -1.76                 |

**Other adverse effects** No information available.

### 13. Disposal considerations

#### Waste treatment methods

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

### 14. Transport information

**TDG** Not regulated

**DOT** Not regulated

**MEX** Not regulated

**IATA** Not regulated

**IMDG** Not regulated

### 15. Regulatory information

#### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

#### International Inventories

Contact supplier for inventory compliance status

**16. Other information**

|                                                                                  |                           |                       |                           |                                           |
|----------------------------------------------------------------------------------|---------------------------|-----------------------|---------------------------|-------------------------------------------|
| <b>NFPA</b>                                                                      | <b>Health hazards</b> 3   | <b>Flammability</b> 0 | <b>Instability</b> 0      | <b>Physical and chemical properties</b> - |
| <b>HMIS</b>                                                                      | <b>Health hazards</b> 3 * | <b>Flammability</b> 0 | <b>Physical hazards</b> 0 | <b>Personal protection</b> X              |
| <i>Chronic Hazard Star Legend                      * = Chronic Health Hazard</i> |                           |                       |                           |                                           |

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
EPA (Environmental Protection Agency)  
Acute Exposure Guideline Level(s) (AEGL(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
Japan GHS Classification  
Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
Organization for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.

**Revision date** 23-Dec-2020

**Revision Note** \*\*\* Indicates this information has changed since the previous revision.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**





# SAFETY DATA SHEET

Revision date 23-Dec-2020

Revision Number 1

## 1. Identification

### Product identifier

**Product Name** SEQuoia Complete Reagent B End Repair Enzyme

### Other means of identification

**Catalog Number(s)** 12011303, 12011945

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

**Restrictions on use** No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories (Canada) Ltd.  
1329 Meyerside Drive  
Mississauga, ON L5T 1C9  
Canada

#### **Technical Service**

1-800-361-1808  
support@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC Canada:1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

Not classified

### Label elements

#### **Hazard statements**

Not classified.

### Other information

Harmful to aquatic life

## 3. Composition/information on ingredients

**Substance**

Not applicable.

**Mixture**

| Chemical name      | CAS No  | Weight-% | Hazardous Material Information Review Act registry number (HMIRA registry #) | Date HMIRA filed and date exemption granted (if applicable) |
|--------------------|---------|----------|------------------------------------------------------------------------------|-------------------------------------------------------------|
| 1,2,3-Propanetriol | 56-81-5 | 30 - 60  | -                                                                            |                                                             |

**4. First-aid measures****Description of first aid measures**

|                |                                                                                                                      |
|----------------|----------------------------------------------------------------------------------------------------------------------|
| General advice | No hazards which require special first aid measures.                                                                 |
| Inhalation     | Remove to fresh air.                                                                                                 |
| Eye contact    | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician. |
| Skin contact   | Wash skin with soap and water.                                                                                       |
| Ingestion      | Rinse mouth thoroughly with water.                                                                                   |

**Most important symptoms and effects, both acute and delayed**

|          |                           |
|----------|---------------------------|
| Symptoms | No information available. |
|----------|---------------------------|

**Indication of any immediate medical attention and special treatment needed**

|                    |                        |
|--------------------|------------------------|
| Note to physicians | Treat symptomatically. |
|--------------------|------------------------|

**5. Fire-fighting measures**

|                                                |                                                                                                                                    |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media                   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                            |
| Unsuitable extinguishing media                 | No information available.                                                                                                          |
| Specific hazards arising from the chemical     | None known.                                                                                                                        |
| Explosion data                                 |                                                                                                                                    |
| Sensitivity to mechanical impact               | None.                                                                                                                              |
| Sensitivity to static discharge                | None.                                                                                                                              |
| Special protective equipment for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment. |

**6. Accidental release measures****Personal precautions, protective equipment and emergency procedures**

|                      |                                     |
|----------------------|-------------------------------------|
| Personal precautions | See section 8 for more information. |
|----------------------|-------------------------------------|

**Methods and material for containment and cleaning up**

Methods for containment Prevent further leakage or spillage if safe to do so.

Methods for cleaning up Pick up and transfer to properly labeled containers.

**7. Handling and storage****Precautions for safe handling**

Advice on safe handling Handle in accordance with good industrial hygiene and safety practice.

**Conditions for safe storage, including any incompatibilities**

Storage Conditions Store according to product and label instructions.

**8. Exposure controls/personal protection****Control parameters****Exposure Limits**

| Chemical name                 | Alberta                   | British Columbia                                      | Ontario | Quebec                    |
|-------------------------------|---------------------------|-------------------------------------------------------|---------|---------------------------|
| 1,2,3-Propanetriol<br>56-81-5 | TWA: 10 mg/m <sup>3</sup> | TWA: 10 mg/m <sup>3</sup><br>TWA: 3 mg/m <sup>3</sup> |         | TWA: 10 mg/m <sup>3</sup> |

**Appropriate engineering controls**

Engineering controls Showers  
Eyewash stations  
Ventilation systems.

**Individual protection measures, such as personal protective equipment**

Eye/face protection Wear safety glasses with side shields (or goggles).

Hand protection Wear suitable gloves.

Skin and body protection Wear suitable protective clothing.

Respiratory protection No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

General hygiene considerations Handle in accordance with good industrial hygiene and safety practice.

**9. Physical and chemical properties****Information on basic physical and chemical properties**

Physical state Liquid  
Appearance aqueous solution  
Color colorless  
Odor Sulfurous  
Odor threshold No information available

**Property**  
pH

**Values**  
7-8

**Remarks • Method**

|                                        |                   |            |
|----------------------------------------|-------------------|------------|
| Melting point / freezing point         | No data available | None known |
| Boiling point / boiling range          | > 100 °C / 212 °F | None known |
| Flash point                            | 160 °C / 320 °F   | None known |
| Evaporation rate                       | No data available | None known |
| Flammability (solid, gas)              | No data available | None known |
| Flammability Limit in Air              |                   | None known |
| Upper flammability or explosive limits | No data available |            |
| Lower flammability or explosive limits | No data available |            |
| Vapor pressure                         | No data available | None known |
| Vapor density                          | No data available | None known |
| Relative density                       | No data available | None known |
| Water solubility                       | Miscible in water |            |
| Solubility in other solvents           | No data available | None known |
| Partition coefficient                  | No data available | None known |
| Autoignition temperature               | No data available | None known |
| Decomposition temperature              |                   | None known |
| Kinematic viscosity                    | No data available | None known |
| Dynamic viscosity                      | No data available | None known |
| <b>Other information</b>               |                   |            |
| Explosive properties                   | Not applicable.   |            |
| Oxidizing properties                   | Not applicable.   |            |
| Softening point                        | Not applicable    |            |
| Molecular weight                       | Not applicable    |            |
| VOC Content (%)                        | Not applicable    |            |

## 10. Stability and reactivity

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Reactivity                         | No information available.                 |
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |
| Hazardous decomposition products   | None known based on information supplied. |

## 11. Toxicological information

### Information on likely routes of exposure

#### Product Information

|              |                                                                   |
|--------------|-------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available. |
| Eye contact  | Specific test data for the substance or mixture is not available. |
| Skin contact | Specific test data for the substance or mixture is not available. |
| Ingestion    | Specific test data for the substance or mixture is not available. |

### Symptoms related to the physical, chemical and toxicological characteristics

|          |                           |
|----------|---------------------------|
| Symptoms | No information available. |
|----------|---------------------------|

### Acute toxicity

#### Numerical measures of toxicity

The following values are calculated based on chapter 3.1 of the GHS document .

ATEmix (oral) 21,000.40 mg/kg  
ATEmix (dermal) 16,683.60 mg/kg

#### Component Information

| Chemical name                 | Oral LD50             | Dermal LD50          | Inhalation LC50                     |
|-------------------------------|-----------------------|----------------------|-------------------------------------|
| 1,2,3-Propanetriol<br>56-81-5 | = 12600 mg/kg ( Rat ) | > 10 g/kg ( Rabbit ) | > 570 mg/m <sup>3</sup> ( Rat ) 1 h |

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Skin corrosion/irritation** Based on available data, the classification criteria are not met.

**Serious eye damage/eye irritation** Based on available data, the classification criteria are not met.

**Respiratory or skin sensitization** Based on available data, the classification criteria are not met.

**Germ cell mutagenicity** Based on available data, the classification criteria are not met.

**Carcinogenicity** Based on available data, the classification criteria are not met.

**Reproductive toxicity** Based on available data, the classification criteria are not met.

**STOT - single exposure** Based on available data, the classification criteria are not met.

**STOT - repeated exposure** Based on available data, the classification criteria are not met.

**Target organ effects** Kidney, Respiratory system, Eyes, Skin.

**Aspiration hazard** Based on available data, the classification criteria are not met.

## 12. Ecological information

**Ecotoxicity** Harmful to aquatic life.

| Chemical name                 | Algae/aquatic plants | Fish                                         | Toxicity to microorganisms | Crustacea                           |
|-------------------------------|----------------------|----------------------------------------------|----------------------------|-------------------------------------|
| 1,2,3-Propanetriol<br>56-81-5 | -                    | LC50: 51 - 57mL/L (96h, Oncorhynchus mykiss) | -                          | EC50: >500mg/L (24h, Daphnia magna) |

**Persistence and degradability** No information available.

**Bioaccumulation** There is no data for this product.

#### Component Information

| Chemical name                 | Partition coefficient |
|-------------------------------|-----------------------|
| 1,2,3-Propanetriol<br>56-81-5 | -1.76                 |

**Other adverse effects** No information available.

## 13. Disposal considerations

#### Waste treatment methods

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

Contaminated packaging Do not reuse empty containers.

## 14. Transport information

|             |               |
|-------------|---------------|
| <u>TDG</u>  | Not regulated |
| <u>DOT</u>  | Not regulated |
| <u>MEX</u>  | Not regulated |
| <u>IATA</u> | Not regulated |
| <u>IMDG</u> | Not regulated |

## 15. Regulatory information

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

### International Regulations

The Montreal Protocol on Substances that Deplete the Ozone Layer Not applicable

The Stockholm Convention on Persistent Organic Pollutants Not applicable

The Rotterdam Convention Not applicable

### International Inventories

Contact supplier for inventory compliance status

## 16. Other information

|             |                  |                |                    |                                                             |
|-------------|------------------|----------------|--------------------|-------------------------------------------------------------|
| <u>NFPA</u> | Health hazards 0 | Flammability 1 | Instability 0      | Physical and chemical properties -<br>Personal protection X |
| <u>HMIS</u> | Health hazards 0 | Flammability 1 | Physical hazards 0 |                                                             |

### Key or legend to abbreviations and acronyms used in the safety data sheet

#### **Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |

#### **Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 EPA (Environmental Protection Agency)  
 Acute Exposure Guideline Level(s) (AELG(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 Japan GHS Classification  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)

National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
Organization for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.

**Revision date** 23-Dec-2020

**Revision Note** \*\*\* Indicates this information has changed since the previous revision.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**



# SAFETY DATA SHEET

Revision date 23-Dec-2020

Revision Number 1

## 1. Identification

### Product identifier

**Product Name** SEQuoia Complete Reagent A Fragmentation Buffer

### Other means of identification

**Catalog Number(s)** 12011302, 12011943

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

**Restrictions on use** No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories (Canada) Ltd.  
1329 Meyerside Drive  
Mississauga, ON L5T 1C9  
Canada

#### **Technical Service**

1-800-361-1808  
support@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC Canada:1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

Not classified

### Label elements

#### **Hazard statements**

Not classified.

### Other information

## 3. Composition/information on ingredients

### Substance



Not applicable.

#### Mixture

The product contains no substances which at their given concentration, are considered to be hazardous to health

### 4. First-aid measures

#### Description of first aid measures

|                |                                                                                                                      |
|----------------|----------------------------------------------------------------------------------------------------------------------|
| General advice | No hazards which require special first aid measures.                                                                 |
| Inhalation     | Remove to fresh air.                                                                                                 |
| Eye contact    | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician. |
| Skin contact   | Wash skin with soap and water.                                                                                       |
| Ingestion      | Rinse mouth thoroughly with water.                                                                                   |

#### Most important symptoms and effects, both acute and delayed

|          |                           |
|----------|---------------------------|
| Symptoms | No information available. |
|----------|---------------------------|

#### Indication of any immediate medical attention and special treatment needed

|                    |                        |
|--------------------|------------------------|
| Note to physicians | Treat symptomatically. |
|--------------------|------------------------|

### 5. Fire-fighting measures

|                                                |                                                                                                                                    |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media                   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                            |
| Unsuitable extinguishing media                 | No information available.                                                                                                          |
| Specific hazards arising from the chemical     | None known.                                                                                                                        |
| Explosion data                                 |                                                                                                                                    |
| Sensitivity to mechanical impact               | None.                                                                                                                              |
| Sensitivity to static discharge                | None.                                                                                                                              |
| Special protective equipment for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment. |

### 6. Accidental release measures

#### Personal precautions, protective equipment and emergency procedures

|                      |                                     |
|----------------------|-------------------------------------|
| Personal precautions | See section 8 for more information. |
|----------------------|-------------------------------------|

#### Methods and material for containment and cleaning up

|                         |                                                       |
|-------------------------|-------------------------------------------------------|
| Methods for containment | Prevent further leakage or spillage if safe to do so. |
| Methods for cleaning up | Pick up and transfer to properly labeled containers.  |

## 7. Handling and storage

### Precautions for safe handling

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

### Conditions for safe storage, including any incompatibilities

**Storage Conditions** Store according to product and label instructions.

## 8. Exposure controls/personal protection

### Control parameters

**Exposure Limits** This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

### Appropriate engineering controls

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

### Individual protection measures, such as personal protective equipment

**Eye/face protection** Wear safety glasses with side shields (or goggles).

**Hand protection** Wear suitable gloves.

**Skin and body protection** Wear suitable protective clothing.

**Respiratory protection** No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**General hygiene considerations** Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

### Information on basic physical and chemical properties

**Physical state** Liquid  
**Appearance** aqueous solution  
**Color** colorless  
**Odor** Sulfurous  
**Odor threshold** No information available

| <u>Property</u>                        | <u>Values</u>     | <u>Remarks • Method</u> |
|----------------------------------------|-------------------|-------------------------|
| pH                                     | 7                 |                         |
| Melting point / freezing point         | No data available | None known              |
| Boiling point / boiling range          | 100 °C / 212 °F   | None known              |
| Flash point                            | No data available | None known              |
| Evaporation rate                       | No data available | None known              |
| Flammability (solid, gas)              | No data available | None known              |
| Flammability Limit in Air              |                   | None known              |
| Upper flammability or explosive limits | No data available |                         |

|                                        |                   |            |
|----------------------------------------|-------------------|------------|
| Lower flammability or explosive limits | No data available |            |
| Vapor pressure                         | No data available | None known |
| Vapor density                          | No data available | None known |
| Relative density                       | No data available | None known |
| Water solubility                       | Miscible in water |            |
| Solubility in other solvents           | No data available | None known |
| Partition coefficient                  | No data available | None known |
| Autoignition temperature               | No data available | None known |
| Decomposition temperature              |                   | None known |
| Kinematic viscosity                    | No data available | None known |
| Dynamic viscosity                      | No data available | None known |
| <b><u>Other information</u></b>        |                   |            |
| Explosive properties                   | Not applicable.   |            |
| Oxidizing properties                   | Not applicable.   |            |
| Softening point                        | Not applicable    |            |
| Molecular weight                       | Not applicable    |            |
| VOC Content (%)                        | Not applicable    |            |

## 10. Stability and reactivity

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Reactivity                         | No information available.                 |
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |
| Hazardous decomposition products   | None known based on information supplied. |

## 11. Toxicological information

### Information on likely routes of exposure

#### Product Information

|              |                                                                   |
|--------------|-------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available. |
| Eye contact  | Specific test data for the substance or mixture is not available. |
| Skin contact | Specific test data for the substance or mixture is not available. |
| Ingestion    | Specific test data for the substance or mixture is not available. |

### Symptoms related to the physical, chemical and toxicological characteristics

|          |                           |
|----------|---------------------------|
| Symptoms | No information available. |
|----------|---------------------------|

### Acute toxicity

#### Numerical measures of toxicity

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                           |                                                                   |
|---------------------------|-------------------------------------------------------------------|
| Skin corrosion/irritation | Based on available data, the classification criteria are not met. |
|---------------------------|-------------------------------------------------------------------|

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Serious eye damage/eye irritation | Based on available data, the classification criteria are not met. |
| Respiratory or skin sensitization | Based on available data, the classification criteria are not met. |
| Germ cell mutagenicity            | Based on available data, the classification criteria are not met. |
| Carcinogenicity                   | Based on available data, the classification criteria are not met. |
| Reproductive toxicity             | Based on available data, the classification criteria are not met. |
| STOT - single exposure            | Based on available data, the classification criteria are not met. |
| STOT - repeated exposure          | Based on available data, the classification criteria are not met. |
| Aspiration hazard                 | Based on available data, the classification criteria are not met. |

## 12. Ecological information

|                               |                           |
|-------------------------------|---------------------------|
| Ecotoxicity                   | .                         |
| Persistence and degradability | No information available. |
| Bioaccumulation               | No information available. |
| Other adverse effects         | No information available. |

## 13. Disposal considerations

### Waste treatment methods

|                                     |                                                                                                                 |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Waste from residues/unused products | Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation. |
| Contaminated packaging              | Do not reuse empty containers.                                                                                  |

## 14. Transport information

|             |               |
|-------------|---------------|
| <u>TDG</u>  | Not regulated |
| <u>DOT</u>  | Not regulated |
| <u>MEX</u>  | Not regulated |
| <u>IATA</u> | Not regulated |
| <u>IMDG</u> | Not regulated |

## 15. Regulatory information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Regulations

The Montreal Protocol on Substances that Deplete the Ozone Layer Not applicable

The Stockholm Convention on Persistent Organic Pollutants Not applicable

The Rotterdam Convention Not applicable

**International Inventories**

Contact supplier for inventory compliance status

## 16. Other information

|             |                  |                |                    |                                    |
|-------------|------------------|----------------|--------------------|------------------------------------|
| <b>NFPA</b> | Health hazards 0 | Flammability 0 | Instability 0      | Physical and chemical properties - |
| <b>HMIS</b> | Health hazards 0 | Flammability 0 | Physical hazards 0 | Personal protection X              |

**Key or legend to abbreviations and acronyms used in the safety data sheet**

**Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
EPA (Environmental Protection Agency)  
Acute Exposure Guideline Level(s) (AEGL(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
Japan GHS Classification  
Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
Organization for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.

**Revision date** 23-Dec-2020

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**End of Safety Data Sheet**



# SAFETY DATA SHEET

Revision date 23-Dec-2020

Revision Number 1

## 1. Identification

### Product identifier

**Product Name** SEQuoia Complete Reagent C Poly(A) Buffer

### Other means of identification

**Catalog Number(s)** 12011304, 12011946

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

**Restrictions on use** No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories (Canada) Ltd.  
1329 Meyerside Drive  
Mississauga, ON L5T 1C9  
Canada

#### **Technical Service**

1-800-361-1808  
support@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC Canada:1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

Not classified

### Label elements

#### **Hazard statements**

Not classified.

### Other information

## 3. Composition/information on ingredients

### Substance

Not applicable.

#### Mixture

The product contains no substances which at their given concentration, are considered to be hazardous to health

### 4. First-aid measures

#### Description of first aid measures

|                |                                                                                                                      |
|----------------|----------------------------------------------------------------------------------------------------------------------|
| General advice | No hazards which require special first aid measures.                                                                 |
| Inhalation     | Remove to fresh air.                                                                                                 |
| Eye contact    | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician. |
| Skin contact   | Wash skin with soap and water.                                                                                       |
| Ingestion      | Rinse mouth thoroughly with water.                                                                                   |

#### Most important symptoms and effects, both acute and delayed

|          |                           |
|----------|---------------------------|
| Symptoms | No information available. |
|----------|---------------------------|

#### Indication of any immediate medical attention and special treatment needed

|                    |                        |
|--------------------|------------------------|
| Note to physicians | Treat symptomatically. |
|--------------------|------------------------|

### 5. Fire-fighting measures

|                                                |                                                                                                                                    |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media                   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                            |
| Unsuitable extinguishing media                 | No information available.                                                                                                          |
| Specific hazards arising from the chemical     | None known.                                                                                                                        |
| Explosion data                                 |                                                                                                                                    |
| Sensitivity to mechanical impact               | None.                                                                                                                              |
| Sensitivity to static discharge                | None.                                                                                                                              |
| Special protective equipment for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment. |

### 6. Accidental release measures

#### Personal precautions, protective equipment and emergency procedures

|                      |                                     |
|----------------------|-------------------------------------|
| Personal precautions | See section 8 for more information. |
|----------------------|-------------------------------------|

#### Methods and material for containment and cleaning up

|                         |                                                       |
|-------------------------|-------------------------------------------------------|
| Methods for containment | Prevent further leakage or spillage if safe to do so. |
| Methods for cleaning up | Pick up and transfer to properly labeled containers.  |

## 7. Handling and storage

### Precautions for safe handling

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

### Conditions for safe storage, including any incompatibilities

**Storage Conditions** Store according to product and label instructions.

## 8. Exposure controls/personal protection

### Control parameters

**Exposure Limits** This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

### Appropriate engineering controls

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

### Individual protection measures, such as personal protective equipment

**Eye/face protection** Wear safety glasses with side shields (or goggles).

**Hand protection** Wear suitable gloves.

**Skin and body protection** Wear suitable protective clothing.

**Respiratory protection** No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**General hygiene considerations** Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

### Information on basic physical and chemical properties

**Physical state** Liquid  
**Appearance** aqueous solution  
**Color** colorless  
**Odor** Sulfurous  
**Odor threshold** No information available

| <u>Property</u>                        | <u>Values</u>     | <u>Remarks • Method</u> |
|----------------------------------------|-------------------|-------------------------|
| pH                                     | 7-8               |                         |
| Melting point / freezing point         | No data available | None known              |
| Boiling point / boiling range          | 100 °C / 212 °F   | None known              |
| Flash point                            | No data available | None known              |
| Evaporation rate                       | No data available | None known              |
| Flammability (solid, gas)              | No data available | None known              |
| Flammability Limit in Air              |                   | None known              |
| Upper flammability or explosive limits | No data available |                         |



|                                        |                   |            |
|----------------------------------------|-------------------|------------|
| Lower flammability or explosive limits | No data available |            |
| Vapor pressure                         | No data available | None known |
| Vapor density                          | No data available | None known |
| Relative density                       | No data available | None known |
| Water solubility                       | Miscible in water |            |
| Solubility in other solvents           | No data available | None known |
| Partition coefficient                  | No data available | None known |
| Autoignition temperature               | No data available | None known |
| Decomposition temperature              |                   | None known |
| Kinematic viscosity                    | No data available | None known |
| Dynamic viscosity                      | No data available | None known |
| <b>Other information</b>               |                   |            |
| Explosive properties                   | Not applicable.   |            |
| Oxidizing properties                   | Not applicable.   |            |
| Softening point                        | Not applicable    |            |
| Molecular weight                       | Not applicable    |            |
| VOC Content (%)                        | Not applicable    |            |

## 10. Stability and reactivity

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Reactivity                         | No information available.                 |
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |
| Hazardous decomposition products   | None known based on information supplied. |

## 11. Toxicological information

### Information on likely routes of exposure

#### Product Information

|              |                                                                   |
|--------------|-------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available. |
| Eye contact  | Specific test data for the substance or mixture is not available. |
| Skin contact | Specific test data for the substance or mixture is not available. |
| Ingestion    | Specific test data for the substance or mixture is not available. |

### Symptoms related to the physical, chemical and toxicological characteristics

|          |                           |
|----------|---------------------------|
| Symptoms | No information available. |
|----------|---------------------------|

### Acute toxicity

#### Numerical measures of toxicity

The following values are calculated based on chapter 3.1 of the GHS document .

|               |                 |
|---------------|-----------------|
| ATEmix (oral) | 60,012.00 mg/kg |
|---------------|-----------------|

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                                          |                                                                   |
|------------------------------------------|-------------------------------------------------------------------|
| <b>Skin corrosion/irritation</b>         | Based on available data, the classification criteria are not met. |
| <b>Serious eye damage/eye irritation</b> | Based on available data, the classification criteria are not met. |
| <b>Respiratory or skin sensitization</b> | Based on available data, the classification criteria are not met. |
| <b>Germ cell mutagenicity</b>            | Based on available data, the classification criteria are not met. |
| <b>Carcinogenicity</b>                   | Based on available data, the classification criteria are not met. |
| <b>Reproductive toxicity</b>             | Based on available data, the classification criteria are not met. |
| <b>STOT - single exposure</b>            | Based on available data, the classification criteria are not met. |
| <b>STOT - repeated exposure</b>          | Based on available data, the classification criteria are not met. |
| <b>Aspiration hazard</b>                 | Based on available data, the classification criteria are not met. |

## 12. Ecological information

|                                      |                           |
|--------------------------------------|---------------------------|
| <b>Ecotoxicity</b>                   | .                         |
| <b>Persistence and degradability</b> | No information available. |
| <b>Bioaccumulation</b>               | No information available. |
| <b>Other adverse effects</b>         | No information available. |

## 13. Disposal considerations

### Waste treatment methods

|                                            |                                                                                                                 |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Waste from residues/unused products</b> | Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation. |
| <b>Contaminated packaging</b>              | Do not reuse empty containers.                                                                                  |

## 14. Transport information

|                    |               |
|--------------------|---------------|
| <b><u>TDG</u></b>  | Not regulated |
| <b><u>DOT</u></b>  | Not regulated |
| <b><u>MEX</u></b>  | Not regulated |
| <b><u>IATA</u></b> | Not regulated |
| <b><u>IMDG</u></b> | Not regulated |

## 15. Regulatory information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Regulations

The Montreal Protocol on Substances that Deplete the Ozone Layer Not applicable

The Stockholm Convention on Persistent Organic Pollutants Not applicable

The Rotterdam Convention Not applicable

#### International Inventories

Contact supplier for inventory compliance status

## 16. Other information

|             |                  |                |                    |                                    |
|-------------|------------------|----------------|--------------------|------------------------------------|
| <u>NFPA</u> | Health hazards 0 | Flammability 0 | Instability 0      | Physical and chemical properties - |
| <u>HMIS</u> | Health hazards 0 | Flammability 0 | Physical hazards 0 | Personal protection X              |

#### Key or legend to abbreviations and acronyms used in the safety data sheet

#### **Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |

#### **Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 EPA (Environmental Protection Agency)  
 Acute Exposure Guideline Level(s) (AEGLS)  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 Japan GHS Classification  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
 Organization for Economic Co-operation and Development Screening Information Data Set  
 RTECS (Registry of Toxic Effects of Chemical Substances)  
 World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.

**Revision date** 23-Dec-2020

**Revision Note** \*\*\* Indicates this information has changed since the previous revision.

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**



# SAFETY DATA SHEET

Revision date 23-Dec-2020

Revision Number 1

## 1. Identification

### Product identifier

**Product Name** SEQuoia Complete Reagent D Poly(A) Polymerase

### Other means of identification

**Catalog Number(s)** 12011291, 12011944

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

**Restrictions on use** No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories (Canada) Ltd.  
1329 Meyerside Drive  
Mississauga, ON L5T 1C9  
Canada

#### **Technical Service**

1-800-361-1808  
support@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC Canada:1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

Not classified

### Label elements

#### **Hazard statements**

Not classified.

### Other information

Harmful to aquatic life

## 3. Composition/information on ingredients

**Substance**

Not applicable.

**Mixture**

| Chemical name | CAS No       | Weight-% | Hazardous Material Information Review Act registry number (HMIRA registry #) | Date HMIRA filed and date exemption granted (if applicable) |
|---------------|--------------|----------|------------------------------------------------------------------------------|-------------------------------------------------------------|
| Trade secret  | Trade secret | 30 - 60  | -                                                                            |                                                             |

**4. First-aid measures****Description of first aid measures**

|                |                                                                                                                      |
|----------------|----------------------------------------------------------------------------------------------------------------------|
| General advice | No hazards which require special first aid measures.                                                                 |
| Inhalation     | Remove to fresh air.                                                                                                 |
| Eye contact    | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician. |
| Skin contact   | Wash skin with soap and water.                                                                                       |
| Ingestion      | Rinse mouth thoroughly with water.                                                                                   |

**Most important symptoms and effects, both acute and delayed**

|          |                           |
|----------|---------------------------|
| Symptoms | No information available. |
|----------|---------------------------|

**Indication of any immediate medical attention and special treatment needed**

|                    |                        |
|--------------------|------------------------|
| Note to physicians | Treat symptomatically. |
|--------------------|------------------------|

**5. Fire-fighting measures**

|                                                |                                                                                                                                    |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media                   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                            |
| Unsuitable extinguishing media                 | No information available.                                                                                                          |
| Specific hazards arising from the chemical     | None known.                                                                                                                        |
| Explosion data                                 |                                                                                                                                    |
| Sensitivity to mechanical impact               | None.                                                                                                                              |
| Sensitivity to static discharge                | None.                                                                                                                              |
| Special protective equipment for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment. |

**6. Accidental release measures****Personal precautions, protective equipment and emergency procedures**

|                      |                                     |
|----------------------|-------------------------------------|
| Personal precautions | See section 8 for more information. |
|----------------------|-------------------------------------|

**Methods and material for containment and cleaning up**

Methods for containment Prevent further leakage or spillage if safe to do so.

Methods for cleaning up Pick up and transfer to properly labeled containers.

**7. Handling and storage****Precautions for safe handling**

Advice on safe handling Handle in accordance with good industrial hygiene and safety practice.

**Conditions for safe storage, including any incompatibilities**

Storage Conditions Store according to product and label instructions.

**8. Exposure controls/personal protection****Control parameters****Exposure Limits**

| Chemical name | Alberta                   | British Columbia                                      | Ontario | Quebec                    |
|---------------|---------------------------|-------------------------------------------------------|---------|---------------------------|
| Trade secret  | TWA: 10 mg/m <sup>3</sup> | TWA: 10 mg/m <sup>3</sup><br>TWA: 3 mg/m <sup>3</sup> |         | TWA: 10 mg/m <sup>3</sup> |

**Appropriate engineering controls**

Engineering controls Showers  
Eyewash stations  
Ventilation systems.

**Individual protection measures, such as personal protective equipment**

Eye/face protection Wear safety glasses with side shields (or goggles).

Hand protection Wear suitable gloves.

Skin and body protection Wear suitable protective clothing.

Respiratory protection No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

General hygiene considerations Handle in accordance with good industrial hygiene and safety practice.

**9. Physical and chemical properties****Information on basic physical and chemical properties**

Physical state Liquid  
Appearance aqueous solution  
Color colorless  
Odor Sulfurous  
Odor threshold No information available

**Property**  
pH

**Values**  
7-8

**Remarks • Method**

|                                        |                   |            |
|----------------------------------------|-------------------|------------|
| Melting point / freezing point         | No data available | None known |
| Boiling point / boiling range          | > 100 °C / 212 °F | None known |
| Flash point                            | 160 °C / 320 °F   | None known |
| Evaporation rate                       | No data available | None known |
| Flammability (solid, gas)              | No data available | None known |
| Flammability Limit in Air              |                   | None known |
| Upper flammability or explosive limits | No data available |            |
| Lower flammability or explosive limits | No data available |            |
| Vapor pressure                         | No data available | None known |
| Vapor density                          | No data available | None known |
| Relative density                       | No data available | None known |
| Water solubility                       | Miscible in water |            |
| Solubility in other solvents           | No data available | None known |
| Partition coefficient                  | No data available | None known |
| Autoignition temperature               | No data available | None known |
| Decomposition temperature              |                   | None known |
| Kinematic viscosity                    | No data available | None known |
| Dynamic viscosity                      | No data available | None known |
| <b>Other information</b>               |                   |            |
| Explosive properties                   | Not applicable.   |            |
| Oxidizing properties                   | Not applicable.   |            |
| Softening point                        | Not applicable    |            |
| Molecular weight                       | Not applicable    |            |
| VOC Content (%)                        | Not applicable    |            |

## 10. Stability and reactivity

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Reactivity                         | No information available.                 |
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |
| Hazardous decomposition products   | None known based on information supplied. |

## 11. Toxicological information

### Information on likely routes of exposure

#### Product Information

|              |                                                                   |
|--------------|-------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available. |
| Eye contact  | Specific test data for the substance or mixture is not available. |
| Skin contact | Specific test data for the substance or mixture is not available. |
| Ingestion    | Specific test data for the substance or mixture is not available. |

### Symptoms related to the physical, chemical and toxicological characteristics

|          |                           |
|----------|---------------------------|
| Symptoms | No information available. |
|----------|---------------------------|

### Acute toxicity

#### Numerical measures of toxicity

The following values are calculated based on chapter 3.1 of the GHS document .

ATEmix (oral) 15,556.60 mg/kg  
ATEmix (dermal) 15,400.50 mg/kg

#### Component Information

| Chemical name | Oral LD50             | Dermal LD50          | Inhalation LC50                     |
|---------------|-----------------------|----------------------|-------------------------------------|
| Trade secret  | = 12600 mg/kg ( Rat ) | > 10 g/kg ( Rabbit ) | > 570 mg/m <sup>3</sup> ( Rat ) 1 h |

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Skin corrosion/irritation** Based on available data, the classification criteria are not met.

**Serious eye damage/eye irritation** Based on available data, the classification criteria are not met.

**Respiratory or skin sensitization** Based on available data, the classification criteria are not met.

**Germ cell mutagenicity** Based on available data, the classification criteria are not met.

**Carcinogenicity** Based on available data, the classification criteria are not met.

**Reproductive toxicity** Based on available data, the classification criteria are not met.

**STOT - single exposure** Based on available data, the classification criteria are not met.

**STOT - repeated exposure** Based on available data, the classification criteria are not met.

**Target organ effects** Kidney, Respiratory system, Eyes, Skin.

**Aspiration hazard** Based on available data, the classification criteria are not met.

## 12. Ecological information

**Ecotoxicity** Harmful to aquatic life.

| Chemical name | Algae/aquatic plants | Fish                                         | Toxicity to microorganisms | Crustacea                           |
|---------------|----------------------|----------------------------------------------|----------------------------|-------------------------------------|
| Trade secret  | -                    | LC50: 51 - 57mL/L (96h, Oncorhynchus mykiss) | -                          | EC50: >500mg/L (24h, Daphnia magna) |

**Persistence and degradability** No information available.

**Bioaccumulation** There is no data for this product.

#### Component Information

| Chemical name | Partition coefficient |
|---------------|-----------------------|
| Trade secret  | -1.76                 |

**Other adverse effects** No information available.

## 13. Disposal considerations

#### Waste treatment methods

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.



Contaminated packaging Do not reuse empty containers.

## 14. Transport information

**TDG** Not regulated  
**DOT** Not regulated  
**MEX** Not regulated  
**IATA** Not regulated  
**IMDG** Not regulated

## 15. Regulatory information

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

### International Regulations

The Montreal Protocol on Substances that Deplete the Ozone Layer Not applicable

The Stockholm Convention on Persistent Organic Pollutants Not applicable

The Rotterdam Convention Not applicable

### International Inventories

Contact supplier for inventory compliance status

## 16. Other information

|             |                  |                |                    |                                                             |
|-------------|------------------|----------------|--------------------|-------------------------------------------------------------|
| <b>NFPA</b> | Health hazards 0 | Flammability 1 | Instability 0      | Physical and chemical properties -<br>Personal protection X |
| <b>HMIS</b> | Health hazards 0 | Flammability 1 | Physical hazards 0 |                                                             |

### Key or legend to abbreviations and acronyms used in the safety data sheet

#### Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |

#### Key literature references and sources for data used to compile the SDS

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 EPA (Environmental Protection Agency)  
 Acute Exposure Guideline Level(s) (AELG(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 Japan GHS Classification  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)

National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
Organization for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.

**Revision date** 23-Dec-2020

**Revision Note** \*\*\* Indicates this information has changed since the previous revision.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**



# SAFETY DATA SHEET

Revision date 23-Dec-2020

Revision Number 1

## 1. Identification

### Product identifier

**Product Name** SEQuoia Complete Reagent E SEQzyme Mix

### Other means of identification

**Catalog Number(s)** 12011292, 12011952

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

**Restrictions on use** No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories (Canada) Ltd.  
1329 Meyerside Drive  
Mississauga, ON L5T 1C9  
Canada

#### **Technical Service**

1-800-361-1808  
support@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC Canada:1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

Not classified

### Label elements

#### **Hazard statements**

Not classified.

### Other information

Harmful to aquatic life

## 3. Composition/information on ingredients

**Substance**

Not applicable.

**Mixture**

| Chemical name      | CAS No  | Weight-% | Hazardous Material Information Review Act registry number (HMIRA registry #) | Date HMIRA filed and date exemption granted (if applicable) |
|--------------------|---------|----------|------------------------------------------------------------------------------|-------------------------------------------------------------|
| 1,2,3-Propanetriol | 56-81-5 | 30 - 60  | -                                                                            |                                                             |

**4. First-aid measures****Description of first aid measures**

|                |                                                                                                                      |
|----------------|----------------------------------------------------------------------------------------------------------------------|
| General advice | No hazards which require special first aid measures.                                                                 |
| Inhalation     | Remove to fresh air.                                                                                                 |
| Eye contact    | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician. |
| Skin contact   | Wash skin with soap and water.                                                                                       |
| Ingestion      | Rinse mouth thoroughly with water.                                                                                   |

**Most important symptoms and effects, both acute and delayed**

|          |                           |
|----------|---------------------------|
| Symptoms | No information available. |
|----------|---------------------------|

**Indication of any immediate medical attention and special treatment needed**

|                    |                        |
|--------------------|------------------------|
| Note to physicians | Treat symptomatically. |
|--------------------|------------------------|

**5. Fire-fighting measures**

|                                                |                                                                                                                                    |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media                   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                            |
| Unsuitable extinguishing media                 | No information available.                                                                                                          |
| Specific hazards arising from the chemical     | None known.                                                                                                                        |
| Explosion data                                 |                                                                                                                                    |
| Sensitivity to mechanical impact               | None.                                                                                                                              |
| Sensitivity to static discharge                | None.                                                                                                                              |
| Special protective equipment for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment. |

**6. Accidental release measures****Personal precautions, protective equipment and emergency procedures**

|                      |                                     |
|----------------------|-------------------------------------|
| Personal precautions | See section 8 for more information. |
|----------------------|-------------------------------------|

**Methods and material for containment and cleaning up**

Methods for containment Prevent further leakage or spillage if safe to do so.

Methods for cleaning up Pick up and transfer to properly labeled containers.

**7. Handling and storage****Precautions for safe handling**

Advice on safe handling Handle in accordance with good industrial hygiene and safety practice.

**Conditions for safe storage, including any incompatibilities**

Storage Conditions Store according to product and label instructions.

**8. Exposure controls/personal protection****Control parameters****Exposure Limits**

| Chemical name                 | Alberta                   | British Columbia                                      | Ontario | Quebec                    |
|-------------------------------|---------------------------|-------------------------------------------------------|---------|---------------------------|
| 1,2,3-Propanetriol<br>56-81-5 | TWA: 10 mg/m <sup>3</sup> | TWA: 10 mg/m <sup>3</sup><br>TWA: 3 mg/m <sup>3</sup> |         | TWA: 10 mg/m <sup>3</sup> |

**Appropriate engineering controls**

Engineering controls Showers  
Eyewash stations  
Ventilation systems.

**Individual protection measures, such as personal protective equipment**

Eye/face protection Wear safety glasses with side shields (or goggles).

Hand protection Wear suitable gloves.

Skin and body protection Wear suitable protective clothing.

Respiratory protection No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

General hygiene considerations Handle in accordance with good industrial hygiene and safety practice.

**9. Physical and chemical properties****Information on basic physical and chemical properties**

Physical state Liquid  
Appearance aqueous solution  
Color colorless  
Odor Odorless  
Odor threshold No information available

**Property**  
pH

**Values**  
7-8

**Remarks • Method**

|                                        |                   |            |
|----------------------------------------|-------------------|------------|
| Melting point / freezing point         | No data available | None known |
| Boiling point / boiling range          | > 100 °C / 212 °F | None known |
| Flash point                            | 160 °C / 320 °F   | None known |
| Evaporation rate                       | No data available | None known |
| Flammability (solid, gas)              | No data available | None known |
| Flammability Limit in Air              |                   | None known |
| Upper flammability or explosive limits | No data available |            |
| Lower flammability or explosive limits | No data available |            |
| Vapor pressure                         | No data available | None known |
| Vapor density                          | No data available | None known |
| Relative density                       | No data available | None known |
| Water solubility                       | Miscible in water |            |
| Solubility in other solvents           | No data available | None known |
| Partition coefficient                  | No data available | None known |
| Autoignition temperature               | No data available | None known |
| Decomposition temperature              |                   | None known |
| Kinematic viscosity                    | No data available | None known |
| Dynamic viscosity                      | No data available | None known |
| <b>Other information</b>               |                   |            |
| Explosive properties                   | Not applicable.   |            |
| Oxidizing properties                   | Not applicable.   |            |
| Softening point                        | Not applicable    |            |
| Molecular weight                       | Not applicable    |            |
| VOC Content (%)                        | Not applicable    |            |

## 10. Stability and reactivity

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Reactivity                         | No information available.                 |
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |
| Hazardous decomposition products   | None known based on information supplied. |

## 11. Toxicological information

### Information on likely routes of exposure

#### Product Information

|              |                                                                   |
|--------------|-------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available. |
| Eye contact  | Specific test data for the substance or mixture is not available. |
| Skin contact | Specific test data for the substance or mixture is not available. |
| Ingestion    | Specific test data for the substance or mixture is not available. |

### Symptoms related to the physical, chemical and toxicological characteristics

|          |                           |
|----------|---------------------------|
| Symptoms | No information available. |
|----------|---------------------------|

### Acute toxicity

#### Numerical measures of toxicity

The following values are calculated based on chapter 3.1 of the GHS document .

ATEmix (oral) 17,747.80 mg/kg  
ATEmix (dermal) 18,200.66 mg/kg

#### Component Information

| Chemical name                 | Oral LD50             | Dermal LD50          | Inhalation LC50                     |
|-------------------------------|-----------------------|----------------------|-------------------------------------|
| 1,2,3-Propanetriol<br>56-81-5 | = 12600 mg/kg ( Rat ) | > 10 g/kg ( Rabbit ) | > 570 mg/m <sup>3</sup> ( Rat ) 1 h |

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Skin corrosion/irritation         | Based on available data, the classification criteria are not met. |
| Serious eye damage/eye irritation | Based on available data, the classification criteria are not met. |
| Respiratory or skin sensitization | Based on available data, the classification criteria are not met. |
| Germ cell mutagenicity            | Based on available data, the classification criteria are not met. |
| Carcinogenicity                   | Based on available data, the classification criteria are not met. |
| Reproductive toxicity             | Based on available data, the classification criteria are not met. |
| STOT - single exposure            | Based on available data, the classification criteria are not met. |
| STOT - repeated exposure          | Based on available data, the classification criteria are not met. |
| Target organ effects              | Kidney, Respiratory system, Eyes, Skin.                           |
| Aspiration hazard                 | Based on available data, the classification criteria are not met. |

## 12. Ecological information

Ecotoxicity Harmful to aquatic life.

| Chemical name                 | Algae/aquatic plants | Fish                                         | Toxicity to microorganisms | Crustacea                           |
|-------------------------------|----------------------|----------------------------------------------|----------------------------|-------------------------------------|
| 1,2,3-Propanetriol<br>56-81-5 | -                    | LC50: 51 - 57mL/L (96h, Oncorhynchus mykiss) | -                          | EC50: >500mg/L (24h, Daphnia magna) |

Persistence and degradability No information available.

Bioaccumulation There is no data for this product.

#### Component Information

| Chemical name                 | Partition coefficient |
|-------------------------------|-----------------------|
| 1,2,3-Propanetriol<br>56-81-5 | -1.76                 |

Other adverse effects No information available.

## 13. Disposal considerations

#### Waste treatment methods

Waste from residues/unused products Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

Contaminated packaging Do not reuse empty containers.

## 14. Transport information

**TDG** Not regulated  
**DOT** Not regulated  
**MEX** Not regulated  
**IATA** Not regulated  
**IMDG** Not regulated

## 15. Regulatory information

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

### International Regulations

The Montreal Protocol on Substances that Deplete the Ozone Layer Not applicable

The Stockholm Convention on Persistent Organic Pollutants Not applicable

The Rotterdam Convention Not applicable

### International Inventories

Contact supplier for inventory compliance status

## 16. Other information

|             |                  |                |                    |                                                             |
|-------------|------------------|----------------|--------------------|-------------------------------------------------------------|
| <b>NFPA</b> | Health hazards 0 | Flammability 1 | Instability 0      | Physical and chemical properties -<br>Personal protection X |
| <b>HMIS</b> | Health hazards 0 | Flammability 1 | Physical hazards 0 |                                                             |

### Key or legend to abbreviations and acronyms used in the safety data sheet

#### Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |

#### Key literature references and sources for data used to compile the SDS

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 EPA (Environmental Protection Agency)  
 Acute Exposure Guideline Level(s) (AELG(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 Japan GHS Classification  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)



National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
Organization for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.

**Revision date** 23-Dec-2020

**Revision Note** \*\*\* Indicates this information has changed since the previous revision.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**



# SAFETY DATA SHEET

Revision date 23-Dec-2020

Revision Number 1

## 1. Identification

### Product identifier

**Product Name** SEQuoia Complete Purification Beads

### Other means of identification

**Catalog Number(s)** 12011294, 12011941, 12012096, 12012097

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

**Restrictions on use** No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories (Canada) Ltd.  
1329 Meyerside Drive  
Mississauga, ON L5T 1C9  
Canada

#### **Technical Service**

1-800-361-1808  
support@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC Canada:1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

Not classified

### Label elements

#### **Hazard statements**

Not classified.

### Other information

Harmful to aquatic life

### 3. Composition/information on ingredients

**Substance**

Not applicable.

**Mixture**

| Chemical name | CAS No       | Weight-% | Hazardous Material Information Review Act registry number (HMIRA registry #) | Date HMIRA filed and date exemption granted (if applicable) |
|---------------|--------------|----------|------------------------------------------------------------------------------|-------------------------------------------------------------|
| Trade secret  | Trade secret | 30 - 60  | -                                                                            |                                                             |

### 4. First-aid measures

**Description of first aid measures**

|                |                                                                                                                      |
|----------------|----------------------------------------------------------------------------------------------------------------------|
| General advice | No hazards which require special first aid measures.                                                                 |
| Inhalation     | Remove to fresh air.                                                                                                 |
| Eye contact    | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician. |
| Skin contact   | Wash skin with soap and water.                                                                                       |
| Ingestion      | Rinse mouth thoroughly with water.                                                                                   |

**Most important symptoms and effects, both acute and delayed**

|          |                           |
|----------|---------------------------|
| Symptoms | No information available. |
|----------|---------------------------|

**Indication of any immediate medical attention and special treatment needed**

|                    |                        |
|--------------------|------------------------|
| Note to physicians | Treat symptomatically. |
|--------------------|------------------------|

### 5. Fire-fighting measures

|                                                |                                                                                                                                    |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media                   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                            |
| Unsuitable extinguishing media                 | No information available.                                                                                                          |
| Specific hazards arising from the chemical     | None known.                                                                                                                        |
| Explosion data                                 |                                                                                                                                    |
| Sensitivity to mechanical impact               | None.                                                                                                                              |
| Sensitivity to static discharge                | None.                                                                                                                              |
| Special protective equipment for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment. |

### 6. Accidental release measures

**Personal precautions, protective equipment and emergency procedures**

**Personal precautions** See section 8 for more information.

#### **Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labeled containers.

## **7. Handling and storage**

#### **Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

#### **Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Store according to product and label instructions.

## **8. Exposure controls/personal protection**

#### **Control parameters**

##### **Exposure Limits**

| Chemical name | Alberta                   | British Columbia                                      | Ontario | Quebec                    |
|---------------|---------------------------|-------------------------------------------------------|---------|---------------------------|
| Trade secret  | TWA: 10 mg/m <sup>3</sup> | TWA: 10 mg/m <sup>3</sup><br>TWA: 3 mg/m <sup>3</sup> |         | TWA: 10 mg/m <sup>3</sup> |

#### **Appropriate engineering controls**

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

#### **Individual protection measures, such as personal protective equipment**

**Eye/face protection** Wear safety glasses with side shields (or goggles).

**Hand protection** Wear suitable gloves.

**Skin and body protection** Wear suitable protective clothing.

**Respiratory protection** No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**General hygiene considerations** Handle in accordance with good industrial hygiene and safety practice.

## **9. Physical and chemical properties**

#### **Information on basic physical and chemical properties**

**Physical state** Liquid  
**Appearance** Suspension  
**Color** colorless  
**Odor** Odorless  
**Odor threshold** No information available

| <u>Property</u>                        | <u>Values</u>       | <u>Remarks • Method</u> |
|----------------------------------------|---------------------|-------------------------|
| pH                                     | 7-8                 |                         |
| Melting point / freezing point         | No data available   | None known              |
| Boiling point / boiling range          | > 100 °C / 212 °F   | None known              |
| Flash point                            | 160 °C / 320 °F     | None known              |
| Evaporation rate                       | No data available   | None known              |
| Flammability (solid, gas)              | No data available   | None known              |
| Flammability Limit in Air              |                     | None known              |
| Upper flammability or explosive limits | No data available   |                         |
| Lower flammability or explosive limits | No data available   |                         |
| Vapor pressure                         | No data available   | None known              |
| Vapor density                          | No data available   | None known              |
| Relative density                       | No data available   | None known              |
| Water solubility                       | Immiscible in water |                         |
| Solubility in other solvents           | No data available   | None known              |
| Partition coefficient                  | No data available   | None known              |
| Autoignition temperature               | No data available   | None known              |
| Decomposition temperature              |                     | None known              |
| Kinematic viscosity                    | No data available   | None known              |
| Dynamic viscosity                      | No data available   | None known              |
| <u>Other information</u>               |                     |                         |
| Explosive properties                   | Not applicable.     |                         |
| Oxidizing properties                   | Not applicable.     |                         |
| Softening point                        | Not applicable      |                         |
| Molecular weight                       | Not applicable      |                         |
| VOC Content (%)                        | Not applicable      |                         |

## 10. Stability and reactivity

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Reactivity                         | No information available.                 |
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |
| Hazardous decomposition products   | None known based on information supplied. |

## 11. Toxicological information

### Information on likely routes of exposure

#### Product Information

|              |                                                                   |
|--------------|-------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available. |
| Eye contact  | Specific test data for the substance or mixture is not available. |
| Skin contact | Specific test data for the substance or mixture is not available. |
| Ingestion    | Specific test data for the substance or mixture is not available. |

### Symptoms related to the physical, chemical and toxicological characteristics

|          |                           |
|----------|---------------------------|
| Symptoms | No information available. |
|----------|---------------------------|

#### Acute toxicity

## Numerical measures of toxicity

The following values are calculated based on chapter 3.1 of the GHS document .

|                 |                 |
|-----------------|-----------------|
| ATEmix (oral)   | 20,655.90 mg/kg |
| ATEmix (dermal) | 22,244.50 mg/kg |

## Component Information

| Chemical name | Oral LD50             | Dermal LD50          | Inhalation LC50                     |
|---------------|-----------------------|----------------------|-------------------------------------|
| Trade secret  | = 12600 mg/kg ( Rat ) | > 10 g/kg ( Rabbit ) | > 570 mg/m <sup>3</sup> ( Rat ) 1 h |

Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Skin corrosion/irritation         | Based on available data, the classification criteria are not met. |
| Serious eye damage/eye irritation | Based on available data, the classification criteria are not met. |
| Respiratory or skin sensitization | Based on available data, the classification criteria are not met. |
| Germ cell mutagenicity            | Based on available data, the classification criteria are not met. |
| Carcinogenicity                   | Based on available data, the classification criteria are not met. |
| Reproductive toxicity             | Based on available data, the classification criteria are not met. |
| STOT - single exposure            | Based on available data, the classification criteria are not met. |
| STOT - repeated exposure          | Based on available data, the classification criteria are not met. |
| Target organ effects              | Kidney, Respiratory system, Eyes, Skin.                           |
| Aspiration hazard                 | Based on available data, the classification criteria are not met. |

**12. Ecological information**

Ecotoxicity Harmful to aquatic life.

| Chemical name | Algae/aquatic plants | Fish                                         | Toxicity to microorganisms | Crustacea                           |
|---------------|----------------------|----------------------------------------------|----------------------------|-------------------------------------|
| Trade secret  | -                    | LC50: 51 - 57mL/L (96h, Oncorhynchus mykiss) | -                          | EC50: >500mg/L (24h, Daphnia magna) |

Persistence and degradability No information available.

Bioaccumulation There is no data for this product.

## Component Information

| Chemical name | Partition coefficient |
|---------------|-----------------------|
| Trade secret  | -1.76                 |

Other adverse effects No information available.

**13. Disposal considerations**Waste treatment methods

|                                            |                                                                                                                 |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Waste from residues/unused products</b> | Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation. |
| <b>Contaminated packaging</b>              | Do not reuse empty containers.                                                                                  |

## 14. Transport information

|                    |               |
|--------------------|---------------|
| <b><u>TDG</u></b>  | Not regulated |
| <b><u>DOT</u></b>  | Not regulated |
| <b><u>MEX</u></b>  | Not regulated |
| <b><u>IATA</u></b> | Not regulated |
| <b><u>IMDG</u></b> | Not regulated |

## 15. Regulatory information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

#### International Inventories

Contact supplier for inventory compliance status

## 16. Other information

|                    |                         |                       |                           |                                           |
|--------------------|-------------------------|-----------------------|---------------------------|-------------------------------------------|
| <b><u>NFPA</u></b> | <b>Health hazards</b> 0 | <b>Flammability</b> 1 | <b>Instability</b> 0      | <b>Physical and chemical properties -</b> |
| <b><u>HMIS</u></b> | <b>Health hazards</b> 0 | <b>Flammability</b> 1 | <b>Physical hazards</b> 0 | <b>Personal protection</b> X              |

### Key or legend to abbreviations and acronyms used in the safety data sheet

#### **Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |

#### **Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 EPA (Environmental Protection Agency)  
 Acute Exposure Guideline Level(s) (AEGL(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 Japan GHS Classification  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)

National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
Organization for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

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**Revision Note** \*\*\* Indicates this information has changed since the previous revision.

**Disclaimer**

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**End of Safety Data Sheet**